PCB Design Report

Infineon REF-FRIDGE-C101T-6ED Motor/Inverter Controller
(Round 2, Problem 2)

Submitted By-

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EDA Tool: **EasyEDA**

Contest: iTest Bangladesh PCB Design Contest 2026, Round Two

Date: 10/10/2026

1. Introduction

This report describes a three-phase inverter and control PCB for a BLDC/PMSM motor, based on the REF-FRIDGE-C101T-6ED architecture. It has an AC input stage with bridge rectifier, an off-line switcher for the control supply, a 6EDL04I06PT gate driver, six IGBTs, and a shunt for current sensing. High-voltage switching is kept apart from low-voltage control circuits.

2. Component List (BOM)

Table 1 lists all components with designator, value, package and source. Parts from the EasyEDA library are marked "Library"; parts built from datasheet dimensions are marked "Custom".

Designator	Component	Value	Package	Source
U2	Off-line switcher LNK3204D	-	SOIC-8 / [package]	[Library / Custom]
U3	Optocoupler PC817	-	SMD-4 or DIP-4	Library
U4	3-phase gate driver 6EDL04I06PT	-	DSO-28	[Library / Custom]
Q3-Q8	IGBT IKD04N60RC2	600 V / 4 A	TO-252 (DPAK)	Library
BR	Bridge rectifier GBU806	8 A / 600 V	GBU	Library
F1	Fuse T3.15A 250V	3.15 A	[holder / SMD]	Library
RV1	Varistor	300 Vrms / 385 Vdc	[disc 14D / 10D]	Library
R14	NTC thermistor 5D-13	5 ohm	Radial THT	Library
L1	Common-mode choke	4 mH	[THT]	[Library / Custom]
RS1	Shunt resistor	150 mOhm, 1%, 3/4 W	2512	Library
J1-J4	Connectors (CON-2PIN, FreqIN, CON-3PIN, TJC2-1)	-	[pitch / package]	Library
TP5, TP6	Test points	-	Pad D1.5 mm	Library
[D, LED]	Diodes and LEDs as in schematic	-	[package]	Library
[R, C, L]	Resistors, capacitors, inductor as in schematic	[values]	[package]	Library

Table 1: Bill of materials

3. Schematic

The schematic has four parts: the AC input stage (F1, RV1, R14, L1, BR), the off-line control supply (U2, U3), the gate driver (U4) with six IGBTs (Q3-Q8), and the shunt sensing (RS1) with connectors J1-J4. Net labels such as DC bus, phase U/V/W and VCC connect all parts.

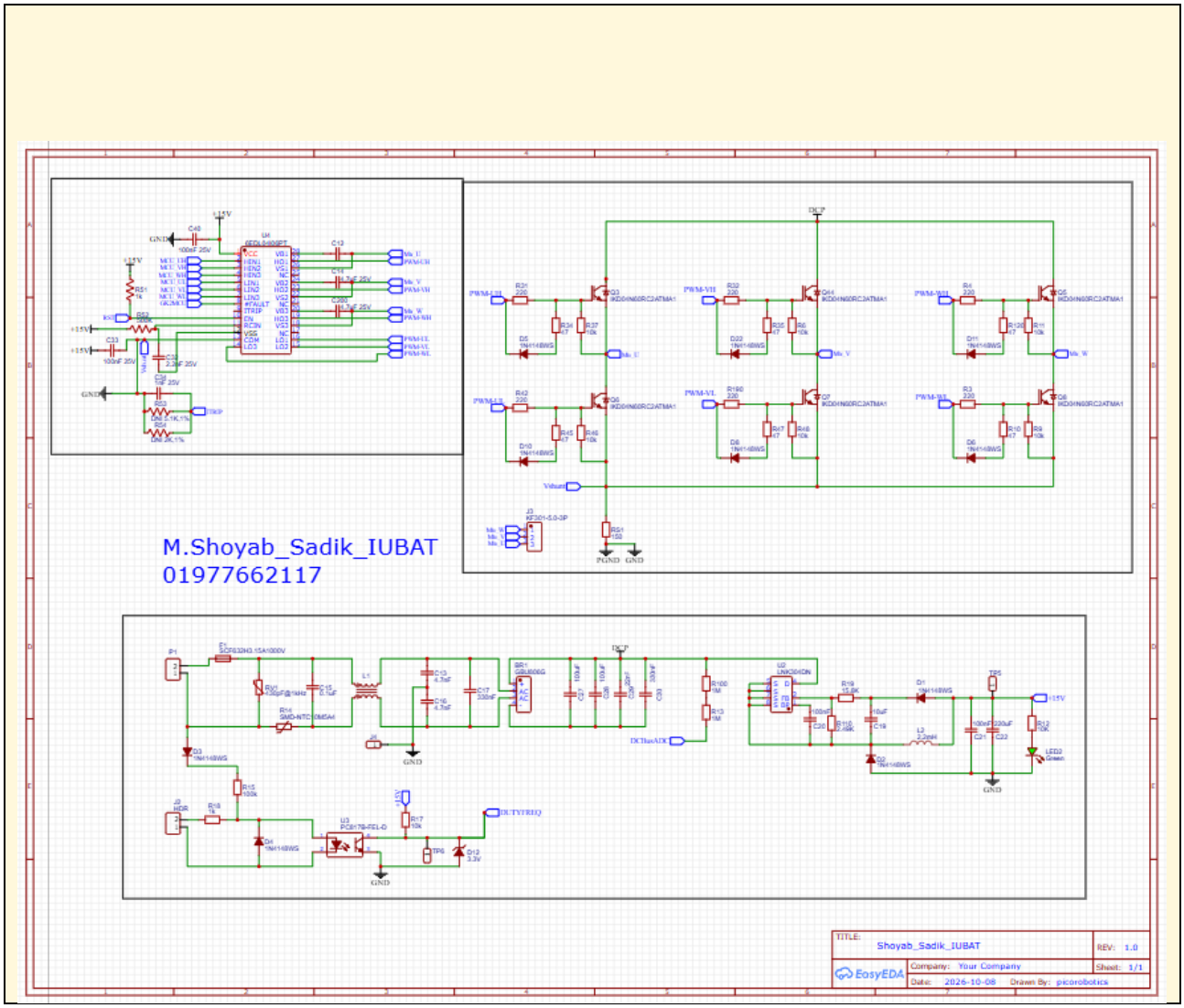


Fig1: Schematic of PCB

4. PCB Design Decisions

The board is [width: 115.443 x height: 54.356] mm with 2 layers. Power flows from the AC input on one side, through the rectifier and DC-link, to the IGBT stage, then to the motor connector. The control section sits on the opposite side.

Requirement	How it was met
Phase stages in logical power flow	[Q3-Q8 in 3 half-bridge pairs between DC bus and motor connector]
Short, direct gate-drive connections	[U4 placed next to Q3-Q8, gate traces X mm]
Small DC-link loop area	[DC bulk/film cap next to IGBTs, bus +/- traces run together]
Switching nodes away from sensing/control	[phase nodes kept X mm from RS1 sense and U3]
Copper sized for current	[AC/DC bus 2.0 mm, phase 1.5 mm, signals 0.25 mm]
Thermal copper and vias	[copper pour and thermal vias under Q3-Q8 and BR]
Creepage and clearance HV to LV	[min X mm between HV and control; slot under U3]
Test points for bus, phase, drive, supply	[TP5, TP6 and added test points with labels]
Silkscreen and warnings	[HV warning, connector names, polarity]

Table 2: Requirements checklist

5. Design Rules

Power traces are wide and short. Control signal traces are 0.25 mm. High-voltage clearance is at least [2.5] mm and control clearance is 0.2 mm. A slot or keep-out separates the high-voltage and control areas.

Rule	Value
Control signal width	0.25 mm
AC input / DC bus width	0.25 mm
Motor phase width	0.25 mm
Clearance (control)	0.3 mm
Clearance / creepage (HV)	0.3 mm
Via (hole / diameter)	0.35 mm
Copper pour	bottom GND for control, copper areas on bus

Table 3: Design rules used

6. PCB Layout Views

Figure 2 shows the PCB design view and Figures 3, 4 and 5 show the 2D top layer, the 2D bottom layer and the 3D view respectively.

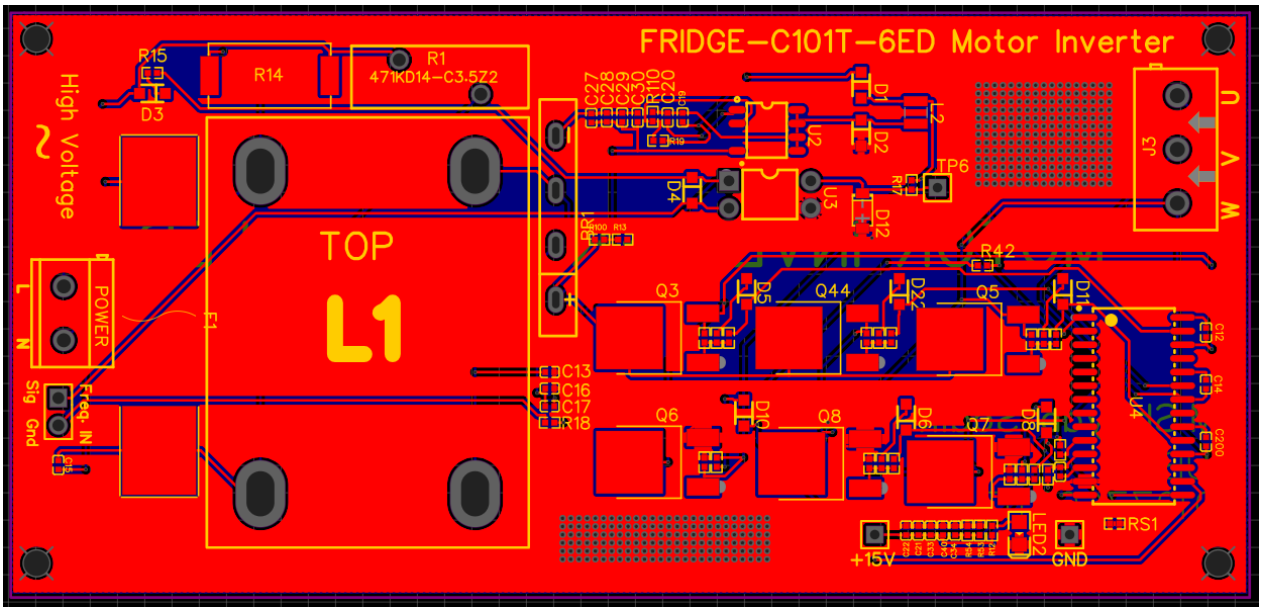


Fig2: PCB Design View

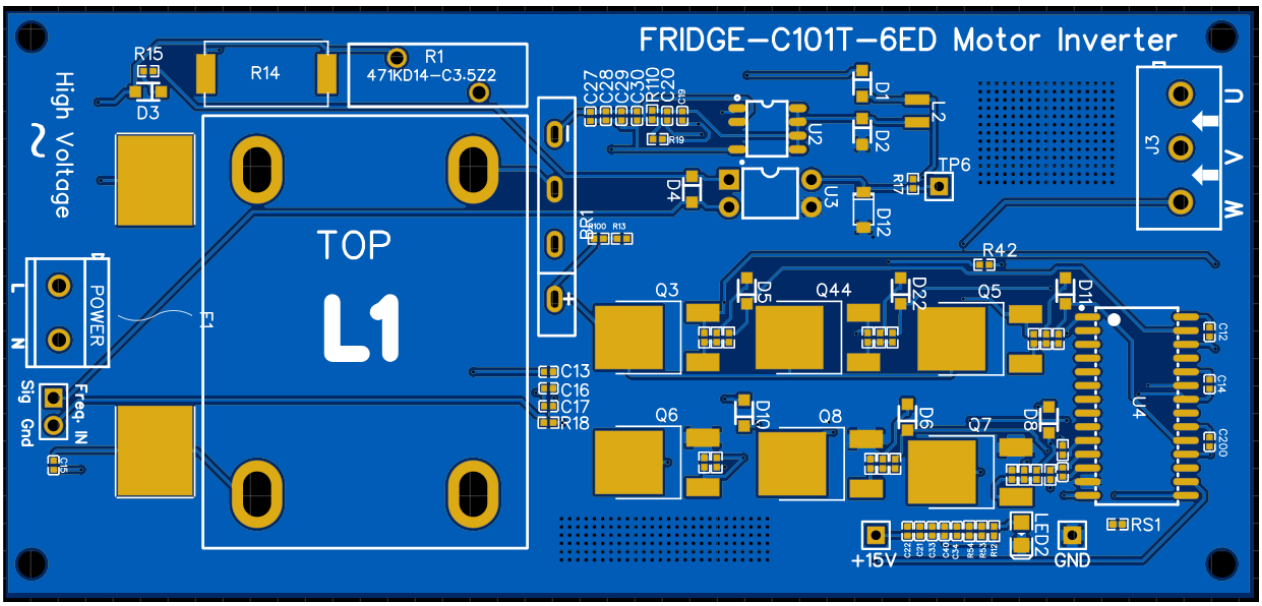


Fig3: Top View of PCB

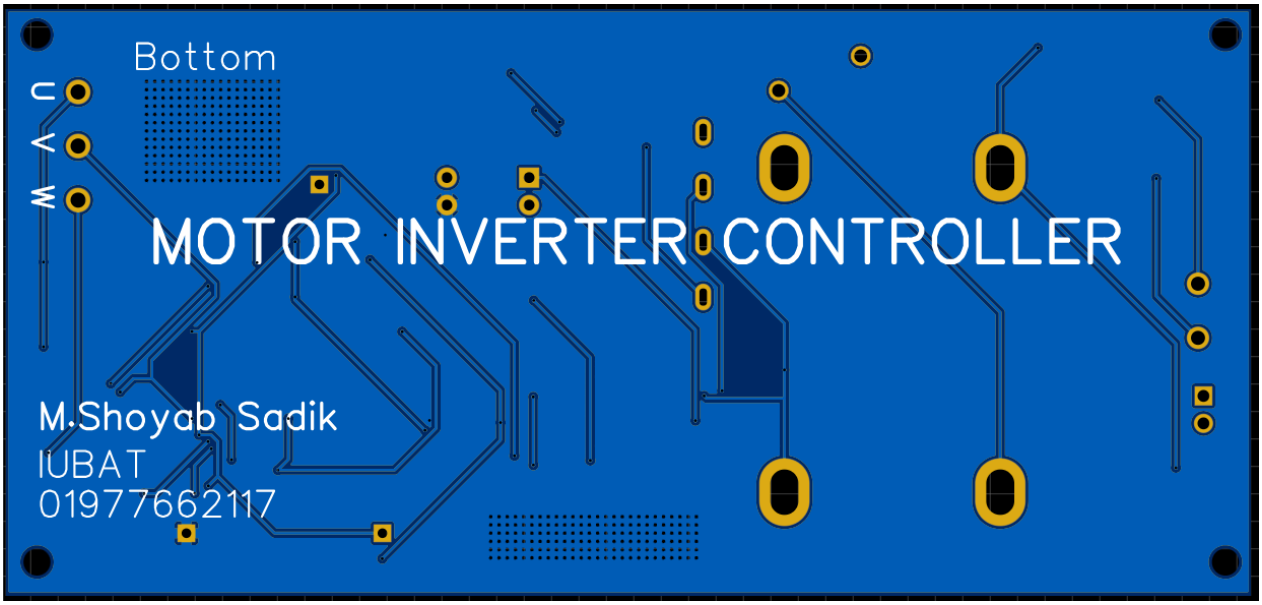


Fig4: Bottom View of PCB

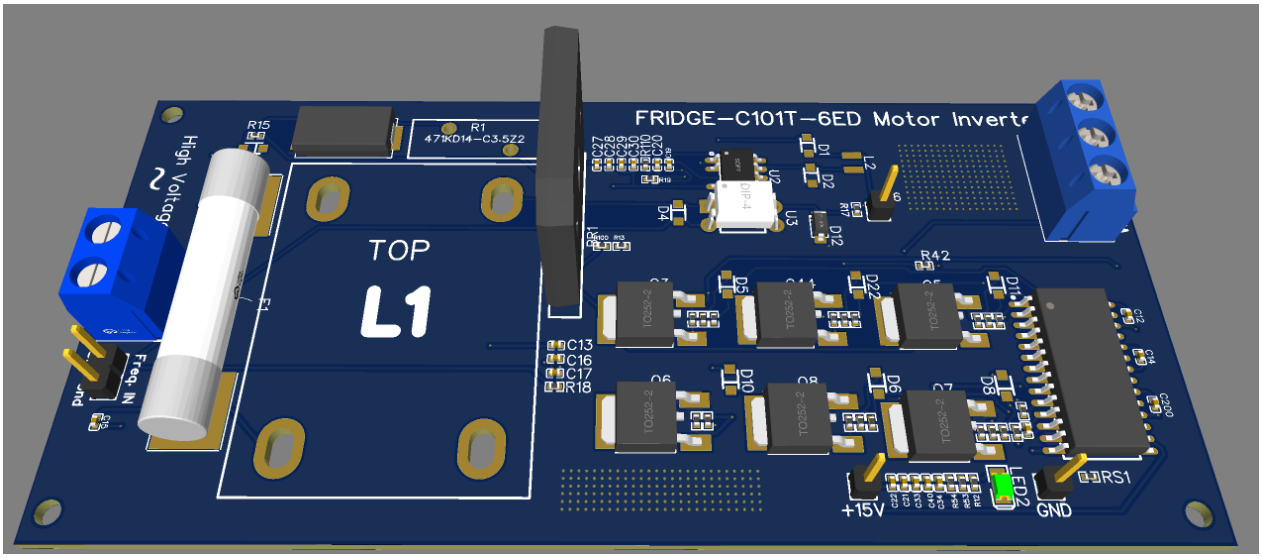


Fig5: 3D View of PCB

7. Conclusion

The design meets the stated requirements: logical power flow, short gate-drive paths, small DC-link loop, safe HV/LV separation, proper copper and thermal vias, useful test points and a clean DRC.

8. Files Submitted

- ShoyabSadik_IUBAT schematic file
- ShoyabSadik_IUBAT board file
- ShoyabSadik_IUBAT pdf (all layers, one per page)
- ShoyabSadik_IUBAT report (this document)

9. Files Source:

- GitHub Link: https://github.com/shoyabsadik/iTest-Bangladesh-PCB-Design-Contest-2026